must take a generator of $C_n(M)$ to twice a generator α of a $\mathbb{Z}$ summand of $C_{n-1}(M)$, in order for $H_n(M;\mathbb{Z}_p)$ to be zero for odd primes p and $\mathbb{Z}_2$ for $p=2$. The cellular chain α must be a cycle since 2α is a boundary and hence a cycle. It follows that the torsion subgroup of $H_{n-1}(M;\mathbb{Z})$ must be a $\mathbb{Z}_2$ generated by α.

Concerning the homology of noncompact manifolds there is the following general statement.

Proposition 3.29. *If M is a connected noncompact n-manifold, then $H_i(M;R)=0$ for $i \ge n$.*

Proof: Represent an element of $H_i(M;R)$ by a cycle z. This has compact image in M, so there is an open set $U \subset M$ containing the image of z and having compact closure $\overline{U} \subset M$. Let $V = M - \overline{U}$. Part of the long exact sequence of the triple $(M, U\cup V, V)$ fits into a commutative diagram

$$\begin{tikzcd}
H_{i+1}(M,U\cup V;R) \arrow[r] & H_i(U\cup V,V;R) \arrow[r] & H_i(M,V;R) \\
& H_i(U;R) \arrow[u, "\approx"'] \arrow[r] & H_i(M;R) \arrow[u]
\end{tikzcd}$$

When $i>n$, the two groups on either side of $H_i(U\cup V,V;R)$ are zero by Lemma 3.27 since $U\cup V$ and V are the complements of compact sets in M. Hence $H_i(U;R)=0$, so z is a boundary in U and therefore in M, and we conclude that $H_i(M;R)=0$.

When $i=n$, the class $[z]\in H_n(M;R)$ defines a section $x\mapsto [z]_x$ of M_R. Since M is connected, this section is determined by its value at a single point, so $[z]_x$ will be zero for all x if it is zero for some x, which it must be since z has compact image and M is noncompact. By Lemma 3.27, z then represents zero in $H_n(M,V;R)$, hence also in $H_n(U;R)$ since the first term in the upper row of the diagram above is zero when $i=n$, by Lemma 3.27 again. So $[z]=0$ in $H_n(M;R)$, and therefore $H_n(M;R)=0$ since $[z]$ was an arbitrary element of this group. $\square$

The Duality Theorem

The form of Poincaré duality we will prove asserts that for an R-orientable closed n-manifold, a certain naturally defined map $H^k(M;R)\to H_{n-k}(M;R)$ is an isomorphism. The definition of this map will be in terms of a more general construction called *cap product*, which has close connections with cup product.

For an arbitrary space X and coefficient ring R, define an R-bilinear cap product $\frown : C_k(X;R)\times C^\ell(X;R)\to C_{k-\ell}(X;R)$ for $k\ge \ell$ by setting

$$\sigma \frown \varphi = \varphi(\sigma\,|\,[v_0,\cdots,v_\ell])\,\sigma\,|\,[v_\ell,\cdots,v_k]$$

for $\sigma:\Delta^k\to X$ and $\varphi\in C^\ell(X;R)$. To see that this induces a cap product in homology and cohomology we use the formula

$$\partial(\sigma \frown \varphi) = (-1)^{\ell}(\partial\sigma \frown \varphi \;-\; \sigma \frown \delta\varphi)$$

which is checked by a calculation:

$$\begin{aligned}
\partial\sigma \frown \varphi &= \sum_{i=0}^{\ell}(-1)^i\varphi(\sigma|[v_0,\cdots,\hat{v}_i,\cdots,v_{\ell+1}])\sigma|[v_{\ell+1},\cdots,v_k] \\
&\qquad + \sum_{i=\ell+1}^{k}(-1)^i\varphi(\sigma|[v_0,\cdots,v_\ell])\sigma|[v_\ell,\cdots,\hat{v}_i,\cdots,v_k] \\
\sigma \frown \delta\varphi &= \sum_{i=0}^{\ell+1}(-1)^i\varphi(\sigma|[v_0,\cdots,\hat{v}_i,\cdots,v_{\ell+1}])\sigma|[v_{\ell+1},\cdots,v_k] \\
\partial(\sigma \frown \varphi) &= \sum_{i=\ell}^{k}(-1)^{i-\ell}\varphi(\sigma|[v_0,\cdots,v_\ell])\sigma|[v_\ell,\cdots,\hat{v}_i,\cdots,v_k]
\end{aligned}$$

From the relation $\partial(\sigma \frown \varphi) = \pm(\partial\sigma \frown \varphi - \sigma \frown \delta\varphi)$ it follows that the cap product of a cycle σ and a cocycle φ is a cycle. Further, if $\partial\sigma = 0$ then $\partial(\sigma \frown \varphi) = \pm(\sigma \frown \delta\varphi)$, so the cap product of a cycle and a coboundary is a boundary. And if $\delta\varphi = 0$ then $\partial(\sigma \frown \varphi) = \pm(\partial\sigma \frown \varphi)$, so the cap product of a boundary and a cocycle is a boundary. These facts imply that there is an induced cap product

$$H_k(X;R) \times H^{\ell}(X;R) \xrightarrow{\ \frown\ } H_{k-\ell}(X;R)$$

which is R-linear in each variable.

Using the same formulas, one checks that cap product has the relative forms

$$\begin{aligned}
H_k(X,A;R) \times H^{\ell}(X;R) &\xrightarrow{\ \frown\ } H_{k-\ell}(X,A;R) \\
H_k(X,A;R) \times H^{\ell}(X,A;R) &\xrightarrow{\ \frown\ } H_{k-\ell}(X;R)
\end{aligned}$$

For example, in the second case the cap product $C_k(X;R) \times C^{\ell}(X;R) \to C_{k-\ell}(X;R)$ restricts to zero on the submodule $C_k(A;R) \times C^{\ell}(X,A;R)$, so there is an induced cap product $C_k(X,A;R) \times C^{\ell}(X,A;R) \to C_{k-\ell}(X;R)$. The formula for $\partial(\sigma \frown \varphi)$ still holds, so we can pass to homology and cohomology groups. There is also a more general relative cap product

$$H_k(X,A \cup B;R) \times H^{\ell}(X,A;R) \xrightarrow{\ \frown\ } H_{k-\ell}(X,B;R),$$

defined when A and B are open sets in X, using the fact that $H_k(X,A \cup B;R)$ can be computed using the chain groups $C_n(X,A+B;R) = C_n(X;R)/C_n(A+B;R)$, as in the derivation of relative Mayer–Vietoris sequences in §2.2.

Cap product satisfies a naturality property that is a little more awkward to state than the corresponding result for cup product since both covariant and contravariant functors are involved. Given a map $f : X \to Y$, the relevant induced maps on homology and cohomology fit into the diagram shown below. It does not quite make sense to say this diagram commutes, but the spirit of commutativity is contained in the formula

$$\begin{tikzcd} H_k(X) \times H^\ell(X) \arrow[r, "\frown"] \arrow[d, "f_*"'] & H_{k-\ell}(X) \arrow[d, "f_*"] \\ H_k(Y) \times H^\ell(Y) \arrow[r, "\frown"] \arrow[u, shift left=8ex, "f^*"'] & H_{k-\ell}(Y) \end{tikzcd}$$

$$f_*(\alpha) \frown \varphi = f_*(\alpha \frown f^*(\varphi))$$

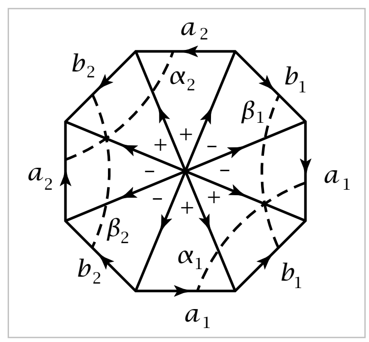
which is obtained by substituting $f\sigma$ for σ in the definition of cap product: $f\sigma \frown \varphi = \varphi(f\sigma\,|\,[v_0,\cdots,v_\ell])\,f\sigma\,|\,[v_\ell,\cdots,v_k]$. There are evident relative versions as well.

Now we can state Poincaré duality for closed manifolds:

Theorem 3.30 (Poincaré Duality). *If M is a closed R-orientable n-manifold with fundamental class $[M] \in H_n(M;R)$, then the map $D: H^k(M;R) \longrightarrow H_{n-k}(M;R)$ defined by $D(\alpha) = [M] \frown \alpha$ is an isomorphism for all k.*

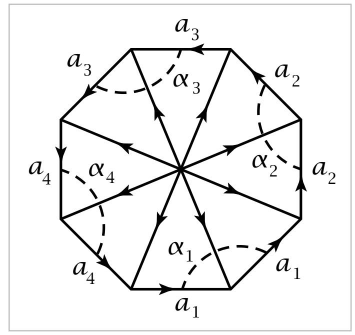
Recall that a fundamental class for M is an element of $H_n(M;R)$ whose image in $H_n(M\,|\,x;R)$ is a generator for each $x \in M$. The existence of such a class was shown in Theorem 3.26.

Example 3.31: Surfaces. Let M be the closed orientable surface of genus g, obtained as usual from a $4g$-gon by identifying pairs of edges according to the word $a_1b_1a_1^{-1}b_1^{-1}\cdots a_gb_ga_g^{-1}b_g^{-1}$. A Δ-complex structure on M is obtained by coning off the $4g$-gon to its center, as indicated in the figure for the case $g = 2$. We can compute cap products using simplicial homology and cohomology since cap products are defined for simplicial homology and cohomology by exactly the same formula as for singular homology and cohomology, so the isomorphism between the simplicial and singular theories respects cap products. A fundamental class $[M]$ generating $H_2(M)$ is represented by the 2-cycle formed by the sum of all $4g$ 2-simplices with the signs indicated.



The edges a_i and b_i form a basis for $H_1(M)$. Under the isomorphism $H^1(M) \approx \mathrm{Hom}(H_1(M),\mathbb{Z})$, the cohomology class α_i corresponding to a_i assigns the value 1 to a_i and 0 to the other basis elements. This class α_i is represented by the cocycle φ_i assigning the value 1 to the 1-simplices meeting the arc labeled α_i in the figure and 0 to the other 1-simplices. Similarly we have a class β_i corresponding to b_i, represented by the cocycle ψ_i assigning the value 1 to the 1-simplices meeting the arc β_i and 0 to the other 1-simplices. Applying the definition of cap product, we have $[M] \frown \varphi_i = b_i$ and $[M] \frown \psi_i = -a_i$ since in both cases there is just one 2-simplex $[v_0,v_1,v_2]$ where φ_i or ψ_i is nonzero on the edge $[v_0,v_1]$. Thus b_i is the Poincaré dual of α_i and $-a_i$ is the Poincaré dual of β_i. If we interpret Poincaré duality entirely in terms of homology, identifying α_i with its Hom-dual a_i and β_i with b_i, then the classes a_i and b_i are Poincaré duals of each other, up to sign at least. Geometrically, Poincaré duality is reflected in the fact that the loops α_i and b_i are homotopic, as are the loops β_i and a_i.

The closed nonorientable surface N of genus g can be treated in the same way if we use $\mathbb{Z}_2$ coefficients. We view N as obtained from a $2g$-gon by identifying consecutive pairs of edges according to the word $a_1^2 \cdots a_g^2$. We have classes $\alpha_i \in H^1(N;\mathbb{Z}_2)$ represented by cocycles φ_i assigning the value 1 to the edges meeting the arc α_i. Then $[N] \frown \varphi_i = a_i$, so a_i is the Poincaré dual of α_i. In terms of homology, a_i is the Hom-dual of α_i, so a_i is its own Poincaré dual. Geometrically, the loops a_i on N are homotopic to their Poincaré dual loops α_i.



Our proof of Poincaré duality, like the construction of fundamental classes, will be by an inductive argument using Mayer–Vietoris sequences. The induction step requires a version of Poincaré duality for open subsets of M, which are noncompact and can satisfy Poincaré duality only when a different kind of cohomology called *cohomology with compact supports* is used.

Cohomology with Compact Supports

Before giving the general definition, let us look at the conceptually simpler notion of simplicial cohomology with compact supports. Here one starts with a Δ-complex X which is locally compact. This is equivalent to saying that every point has a neighborhood that meets only finitely many simplices. Consider the subgroup $\Delta_c^i(X;G)$ of the simplicial cochain group $\Delta^i(X;G)$ consisting of cochains that are compactly supported in the sense that they take nonzero values on only finitely many simplices. The coboundary of such a cochain φ can have a nonzero value only on those $(i+1)$-simplices having a face on which φ is nonzero, and there are only finitely many such simplices by the local compactness assumption, so $\delta\varphi$ lies in $\Delta_c^{i+1}(X;G)$. Thus we have a subcomplex of the simplicial cochain complex. The cohomology groups for this subcomplex will be denoted temporarily by $H_c^i(X;G)$.

Example 3.32. Let us compute these cohomology groups when $X = \mathbb{R}$ with the Δ-complex structure having vertices at the integer points. For a simplicial 0-cochain to be a cocycle it must take the same value on all vertices, but then if the cochain lies in $\Delta_c^0(X)$ it must be identically zero. Thus $H_c^0(\mathbb{R};G) = 0$. However, $H_c^1(\mathbb{R};G)$ is nonzero. Namely, consider the map $\Sigma : \Delta_c^1(\mathbb{R};G) \to G$ sending each cochain to the sum of its values on all the 1-simplices. Note that Σ is not defined on all of $\Delta^1(X)$, just on $\Delta_c^1(X)$. The map Σ vanishes on coboundaries, so it induces a map $H_c^1(\mathbb{R};G) \to G$. This is surjective since every element of $\Delta_c^1(X)$ is a cocycle. It is an easy exercise to verify that it is also injective, so $H_c^1(\mathbb{R};G) \approx G$.

Compactly supported cellular cohomology for a locally compact CW complex could be defined in a similar fashion, using cellular cochains that are nonzero on only finitely many cells. However, what we really need is singular cohomology with compact supports for spaces without any simplicial or cellular structure. The quickest definition of this is the following. Let $C_c^i(X;G)$ be the subgroup of $C^i(X;G)$ consisting of cochains $\varphi : C_i(X) \to G$ for which there exists a compact set $K = K_\varphi \subset X$ such that φ is zero on all chains in $X - K$. Note that $\delta\varphi$ is then also zero on chains in $X - K$, so $\delta\varphi$ lies in $C_c^{i+1}(X;G)$ and the $C_c^i(X;G)$'s for varying i form a subcomplex of the singular cochain complex of X. The cohomology groups $H_c^i(X;G)$ of this subcomplex are the **cohomology groups with compact supports**.

Cochains in $C_c^i(X;G)$ have compact support in only a rather weak sense. A stronger and perhaps more natural condition would have been to require cochains to be nonzero only on singular simplices contained in some compact set, depending on the cochain. However, cochains satisfying this condition do not in general form a subcomplex of the singular cochain complex. For example, if $X = \mathbb{R}$ and φ is a 0-cochain assigning a nonzero value to one point of $\mathbb{R}$ and zero to all other points, then $\delta\varphi$ assigns a nonzero value to arbitrarily large 1-simplices.

It will be quite useful to have an alternative definition of $H_c^i(X;G)$ in terms of algebraic limits, which enter the picture in the following way. The cochain group $C_c^i(X;G)$ is the union of its subgroups $C^i(X, X-K;G)$ as K ranges over compact subsets of X. Each inclusion $K \hookrightarrow L$ induces inclusions $C^i(X,X-K;G) \hookrightarrow C^i(X,X-L;G)$ for all i, so there are induced maps $H^i(X,X-K;G) \to H^i(X,X-L;G)$. These need not be injective, but one might still hope that $H_c^i(X;G)$ is somehow describable in terms of the system of groups $H^i(X,X-K;G)$ for varying K. This is indeed the case, and it is algebraic limits that provide the description.

Suppose one has abelian groups G_α indexed by some partially ordered index set I having the property that for each pair $\alpha,\beta \in I$ there exists $\gamma \in I$ with $\alpha \le \gamma$ and $\beta \le \gamma$. Such an I is called a **directed set**. Suppose also that for each pair $\alpha \le \beta$ one has a homomorphism $f_{\alpha\beta} : G_\alpha \to G_\beta$, such that $f_{\alpha\alpha} = \mathbb{1}$ for each α, and if $\alpha \le \beta \le \gamma$ then $f_{\alpha\gamma}$ is the composition of $f_{\alpha\beta}$ and $f_{\beta\gamma}$. Given this data, which is called a **directed system** of groups, there are two equivalent ways of defining the **direct limit** group $\varinjlim G_\alpha$. The shorter definition is that $\varinjlim G_\alpha$ is the quotient of the direct sum $\bigoplus_\alpha G_\alpha$ by the subgroup generated by all elements of the form $a - f_{\alpha\beta}(a)$ for $a \in G_\alpha$, where we are viewing each G_α as a subgroup of $\bigoplus_\alpha G_\alpha$. The other definition, which is often more convenient to work with, runs as follows. Define an equivalence relation on the set $\coprod_\alpha G_\alpha$ by $a \sim b$ if $f_{\alpha\gamma}(a) = f_{\beta\gamma}(b)$ for some γ, where $a \in G_\alpha$ and $b \in G_\beta$. This is clearly reflexive and symmetric, and transitivity follows from the directed set property. It could also be described as the equivalence relation generated by setting $a \sim f_{\alpha\beta}(a)$. Any two equivalence classes $[a]$ and $[b]$ have representatives a' and b' lying in the same G_γ, so define $[a] + [b] = [a' + b']$. One checks this is well-defined and gives an abelian group structure to the set of equivalence classes. It is easy to check further that the map sending an equivalence class $[a]$ to the coset of a

in $\varinjlim G_\alpha$ is a homomorphism, with an inverse induced by the map $\sum_i a_i \mapsto \sum_i [a_i]$ for $a_i \in G_{\alpha_i}$. Thus we can identify $\varinjlim G_\alpha$ with the group of equivalence classes $[a]$.

A useful consequence of this is that if we have a subset $J \subset I$ with the property that for each $\alpha \in I$ there exists a $\beta \in J$ with $\alpha \le \beta$, then $\varinjlim G_\alpha$ is the same whether we compute it with α varying over I or just over J. In particular, if I has a maximal element γ, we can take $J = \{\gamma\}$ and then $\varinjlim G_\alpha = G_\gamma$.

Suppose now that we have a space X expressed as the union of a collection of subspaces X_α forming a directed set with respect to the inclusion relation. Then the groups $H_i(X_\alpha;G)$ for fixed i and G form a directed system, using the homomorphisms induced by inclusions. The natural maps $H_i(X_\alpha;G) \to H_i(X;G)$ induce a homomorphism $\varinjlim H_i(X_\alpha;G) \to H_i(X;G)$.

Proposition 3.33. *If a space X is the union of a directed set of subspaces X_α with the property that each compact set in X is contained in some X_α, then the natural map $\varinjlim H_i(X_\alpha;G) \to H_i(X;G)$ is an isomorphism for all i and G.*

Proof: For surjectivity, represent a cycle in X by a finite sum of singular simplices. The union of the images of these singular simplices is compact in X, hence lies in some X_α, so the map $\varinjlim H_i(X_\alpha;G) \to H_i(X;G)$ is surjective. Injectivity is similar: If a cycle in some X_α is a boundary in X, compactness implies it is a boundary in some $X_\beta \supset X_\alpha$, hence represents zero in $\varinjlim H_i(X_\alpha;G)$. $\square$

Now we can give the alternative definition of cohomology with compact supports in terms of direct limits. For a space X, the compact subsets $K \subset X$ form a directed set under inclusion since the union of two compact sets is compact. To each compact $K \subset X$ we associate the group $H^i(X, X-K;G)$, with a fixed i and coefficient group G, and to each inclusion $K \subset L$ of compact sets we associate the natural homomorphism $H^i(X, X-K;G) \to H^i(X, X-L;G)$. The resulting limit group $\varinjlim H^i(X, X-K;G)$ is then equal to $H_c^i(X;G)$ since each element of this limit group is represented by a cocycle in $C^i(X, X-K;G)$ for some compact K, and such a cocycle is zero in $\varinjlim H^i(X, X-K;G)$ iff it is the coboundary of a cochain in $C^{i-1}(X, X-L;G)$ for some compact $L \supset K$.

Note that if X is compact, then $H_c^i(X;G) = H^i(X;G)$ since there is a unique maximal compact set $K \subset X$, namely X itself. This is also immediate from the original definition since $C_c^i(X;G) = C^i(X;G)$ if X is compact.

Example 3.34: $H_c^*(\mathbb{R}^n;G)$. To compute $\varinjlim H^i(\mathbb{R}^n, \mathbb{R}^n - K;G)$ it suffices to let K range over balls B_k of integer radius k centered at the origin since every compact set is contained in such a ball. Since $H^i(\mathbb{R}^n, \mathbb{R}^n - B_k;G)$ is nonzero only for $i = n$, when it is G, and the maps $H^n(\mathbb{R}^n, \mathbb{R}^n - B_k;G) \to H^n(\mathbb{R}^n, \mathbb{R}^n - B_{k+1};G)$ are isomorphisms, we deduce that $H_c^i(\mathbb{R}^n;G) = 0$ for $i \ne n$ and $H_c^n(\mathbb{R}^n;G) \approx G$.

This example shows that cohomology with compact supports is not an invariant of homotopy type. This can be traced to difficulties with induced maps. For example, the constant map from $\mathbb{R}^n$ to a point does not induce a map on cohomology with compact supports. The maps which do induce maps on H_c^* are the *proper* maps, those for which the inverse image of each compact set is compact. In the proof of Poincaré duality, however, we will need induced maps of a different sort going in the opposite direction from what is usual for cohomology, maps $H_c^i(U;G) \to H_c^i(V;G)$ associated to inclusions $U \hookrightarrow V$ of open sets in the fixed manifold M.

The group $H^i(X, X-K;G)$ for K compact depends only on a neighborhood of K in X by excision, assuming X is Hausdorff so that K is closed. As convenient shorthand notation we will write this group as $H^i(X\,|\,K;G)$, in analogy with the similar notation used earlier for local homology. One can think of cohomology with compact supports as the limit of these 'local cohomology groups at compact subsets.'

Duality for Noncompact Manifolds

For M an R-orientable n-manifold, possibly noncompact, we can define a duality map $D_M : H_c^k(M;R) \to H_{n-k}(M;R)$ by a limiting process in the following way. For compact sets $K \subset L \subset M$ we have a diagram

$$\begin{array}{ccccc} H_n(M|L;R) & \times & H^k(M|L;R) & \xrightarrow{\frown} & \\ \downarrow i_* & & \uparrow i^* & & H_{n-k}(M;R) \\ H_n(M|K;R) & \times & H^k(M|K;R) & \xrightarrow{\frown} & \end{array}$$

where $H_n(M\,|\,A;R) = H_n(M, M-A;R)$ and $H^k(M\,|\,A;R) = H^k(M, M-A;R)$. By Lemma 3.27 there are unique elements $\mu_K \in H_n(M\,|\,K;R)$ and $\mu_L \in H_n(M\,|\,L;R)$ restricting to a given orientation of M at each point of K and L, respectively. From the uniqueness we have $i_*(\mu_L) = \mu_K$. The naturality of cap product implies that $i_*(\mu_L) \frown x = \mu_L \frown i^*(x)$ for all $x \in H^k(M\,|\,K;R)$, so $\mu_K \frown x = \mu_L \frown i^*(x)$. Therefore, letting K vary over compact sets in M, the homomorphisms $H^k(M\,|\,K;R) \to H_{n-k}(M;R)$, $x \mapsto \mu_K \frown x$, induce in the limit a duality homomorphism $D_M : H_c^k(M;R) \to H_{n-k}(M;R)$.

Since $H_c^*(M;R) = H^*(M;R)$ if M is compact, the following theorem generalizes Poincaré duality for closed manifolds:

Theorem 3.35. *The duality map $D_M : H_c^k(M;R) \to H_{n-k}(M;R)$ is an isomorphism for all k whenever M is an R-oriented n-manifold.*

The proof will not be difficult once we establish a technical result stated in the next lemma, concerning the commutativity of a certain diagram. Commutativity statements of this sort are usually routine to prove, but this one seems to be an exception. The reader who consults other books for alternative expositions will find somewhat uneven treatments of this technical point, and the proof we give is also not as simple as one would like.

The coefficient ring R will be fixed throughout the proof, and for simplicity we will omit it from the notation for homology and cohomology.

Lemma 3.36. *If M is the union of two open sets U and V, then there is a diagram of Mayer–Vietoris sequences, commutative up to sign*:

$$\begin{tikzcd}
\cdots \arrow[r] & H_c^k(U\cap V) \arrow[r] \arrow[d, "D_{U\cap V}"] & H_c^k(U)\oplus H_c^k(V) \arrow[r] \arrow[d, "D_U\oplus -D_V"] & H_c^k(M) \arrow[r] \arrow[d, "D_M"] & H_c^{k+1}(U\cap V) \arrow[r] \arrow[d, "D_{U\cap V}"] & \cdots \\
\cdots \arrow[r] & H_{n-k}(U\cap V) \arrow[r] & H_{n-k}(U)\oplus H_{n-k}(V) \arrow[r] & H_{n-k}(M) \arrow[r] & H_{n-k-1}(U\cap V) \arrow[r] & \cdots
\end{tikzcd}$$

Proof: Compact sets $K\subset U$ and $L\subset V$ give rise to the Mayer–Vietoris sequence in the upper row of the following diagram, whose lower row is also a Mayer–Vietoris sequence.

$$\begin{tikzcd}
\cdots \arrow[r] & H^k(M|K\cap L) \arrow[r] \arrow[d, "\approx"] & H^k(M|K)\oplus H^k(M|L) \arrow[r] \arrow[d, "\approx"] & H^k(M|K\cup L) \arrow[r] \arrow[dd, "\mu_{K\cup L}\frown"] & \cdots \\
& H^k(U\cap V|K\cap L) \arrow[d, "\mu_{K\cap L}\frown"] & H^k(U|K)\oplus H^k(V|L) \arrow[d, "\mu_K\frown\oplus-\mu_L\frown"] & & \\
\cdots \arrow[r] & H_{n-k}(U\cap V) \arrow[r] & H_{n-k}(U)\oplus H_{n-k}(V) \arrow[r] & H_{n-k}(M) \arrow[r] & \cdots
\end{tikzcd}$$

The two maps labeled isomorphisms come from excision. Assuming this diagram commutes, consider passing to the limit over compact sets $K\subset U$ and $L\subset V$. Since each compact set in $U\cap V$ is contained in an intersection $K\cap L$ of compact sets $K\subset U$ and $L\subset V$, and similarly for $U\cup V$, the diagram induces a limit diagram having the form stated in the lemma. The first row of this limit diagram is exact since a direct limit of exact sequences is exact; this is an exercise at the end of the section, and follows easily from the definition of direct limits.

It remains to consider the commutativity of the preceding diagram involving K and L. In the two squares shown, not involving boundary or coboundary maps, it is a triviality to check commutativity at the level of cycles and cocycles. Less trivial is the third square, which we rewrite in the following way:

$$(*)\qquad \begin{tikzcd}
H^k(M|K\cup L) \arrow[r, "\delta"] \arrow[d, "\mu_{K\cup L}\frown"] & H^{k+1}(M|K\cap L) \arrow[r, "\approx"] & H^{k+1}(U\cap V|K\cap L) \arrow[d, "\mu_{K\cap L}\frown"] \\
H_{n-k}(M) \arrow[rr, "\partial"] & & H_{n-k-1}(U\cap V)
\end{tikzcd}$$

Letting $A=M-K$ and $B=M-L$, the map δ is the coboundary map in the Mayer–Vietoris sequence obtained from the short exact sequence of cochain complexes

$$0\longrightarrow C^*(M,A+B)\longrightarrow C^*(M,A)\oplus C^*(M,B)\longrightarrow C^*(M,A\cap B)\longrightarrow 0$$

where $C^*(M,A+B)$ consists of cochains on M vanishing on chains in A and chains in B. To evaluate the Mayer–Vietoris coboundary map δ on a cohomology class represented by a cocycle $\varphi\in C^*(M,A\cap B)$, the first step is to write $\varphi=\varphi_A-\varphi_B$